

Intrusion Detection System Based on Machine Learning Techniques: A Survey

1st Omar I. Sheet
Master student

University of Mosul-
College of Computer Sciences
& Math. – Software Engineering Dept.
omar.21csp1@student.uomosul.edu.iq

2nd Laheeb M. Ibrahim
Professor

University of Mosul-
College of Computer Sciences
& Math. – Software Engineering Dept.
laheeb_alzubaidy321966@uomosul.edu.iq

Abstract— The volume of network traffic data has become so big and complicated as a result of the development in Internet-based services that it is extremely difficult to process using typical data processing techniques. Due to the enormous and complicated nature of network traffic data, fast and effective cybersecurity intrusion detection is a highly difficult task. In order to detect hostile traffic as soon as feasible, a realistic cyber security intrusion detection system should be able to handle a huge volume of network traffic data as quickly as possible.

This paper studies a classification-based intrusion detection algorithms based on Machine Learning for speedy and effective intrusion detection in huge network traffic, including Support Vector Machines (SVM), Random Forests(RF), Decision Trees(DT), Naïve Bayes (NB), Deep Neural Network (DNN), Extreme Gradient Boosting (XG Boost), Non-dominated Sorting Genetic Algorithm (NSGA2), as well as Deep Belief Network (DBN) and artificial neural network(AI). It has been applied to ACTUAL TIME The KDD dataset, KDD Cup 99 dataset, NSL-KDD dataset, CICIDS-2017 dataset, CICIDS-2018 dataset, ISCX Dataset, CICAndMal2017 dataset and UNSW-NB15 to compare performance based on rating Intrusion detection systems which are assessed in terms of training time, prediction time, accuracy.

Keywords: Cyber security, Intrusion Detection, Network Traffic Analysis, Machine Learning (ML), Accuracy.

I. INTRODUCTION

A. Overview

The Internet has become a more popular source of both knowledge and (online) services, where it is a global network comprising millions of interconnected computers, networks and other devices, the advancement of computer systems, networks, and mobile devices that have significantly boosted Internet usage. As a result, the Internet has been a target for cybercriminals and adversaries [1] [2]. A safe and stable computer system must assure information confidentiality, availability, and integrity. The integrity and security of a computer system are jeopardized when an unauthorized human or software joins a computer or network with the internet to damage or disrupt regular operations [3].

Cybersecurity refers to the collection of security measures that may be used to safeguard the cyberspace and user assets from unwanted access and assaults. The primary goal of a cyber-defense system is for data to be confidential, integral, and accessible [4].

ML approaches are being used on both the offensive and defensive sides of the net. On the attacker's side, ML methods are used to breach the defense wall. On the defense side, ML approaches are used to develop quick and robust defense plans. ML approaches are critical in combating cybersecurity risks and assaults such as intrusion detection systems [5],[6], malware detection [7], phishing detection [8],[9], spam detection [10],[11], and fraud detection [12].

An intrusion detection system (IDS) is used to protect computer networks from malicious attacks. The basic classifications of an intrusion detection system for network analysis are signature-based, anomaly-based, and hybrid-based. ML approaches have made significant contributions to detect many sorts of intrusions on network and host systems. [13].

B. Problem Statement

Countries all around the world have become so reliant on cyberspace for communication and control of the real world that it is now difficult to disengage from it. As a result, cyberspace is rapidly affecting each country's security tasks and functions [14]. It is impossible to establish assurances in the product supply chain process due to the worldwide manufacturing of software and hardware items.

Cyber-threats have a very broad range of impacts, therefore we must have a system that can regulate real-world activities. Cyberspace activities, are controlled by a limited number of persons. Users are unable to change or control the software and hardware that they utilize. It is no secret that only a limited number of persons are able to control or manage cyber warfare properly [15]. Hence, a cybersecurity, which is frequently a complicated and tough operation, must be secured from threats that influence it. Machine learning will be used in this paper to see how to safeguard it.[16].

" The study is structured as follows: Section II provides background information on cybersecurity and machine learning, Section III provides a literature review, and Section IV provides a conclusion"

II. BACKGROUND

A comprehensive cybersecurity strategy has numerous levels of protection dispersed throughout the computers,

networks, applications, or data that are to be protected. People, procedures, and technology in a corporation should all work together to provide an effective protection against cyberattacks. A unified threat management system will automate and speed up core security operations processes such as detection, investigation, and remediation.

In addition to the possibility of occurring any possible dangers, cyber security and information security both seek to safeguard data against breaches or assaults, though they are not the same. Information security protects data in a specific location, whereas cyber security protects data in general by shielding it from any external attack on the data.

The term "risk" is commonly used in the literature to denote the likelihood of loss and/or damage. Various sorts of risk may be discovered depending on the circumstances, such as commercial risk, economic risk, and safety risk [16]. In the field of computer science, AI is commonly considered to denote the capacity of any machine, ranging from conventional computers to robots, to execute challenging tasks that typically need human-like intelligence. There are four forms of AI in theory: reactive machines, limited memory machines, theory of mind, and self-aware AI.[16]

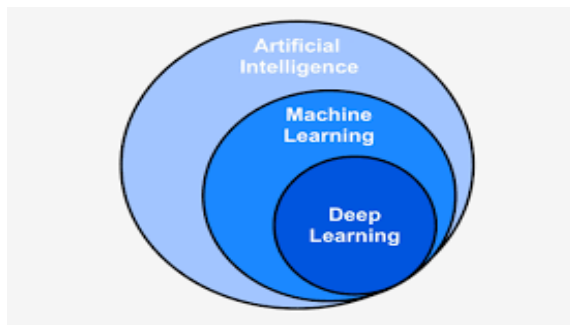


Fig. 1: Artificial Intelligence overview [16]

In the literature, AI is often referred to as ML and deep learning (DL). Figure 1 depicts a high-level overview of the AI, where AI and ML are frequently used interchangeably in practice. In general, AI is defined as the science of designing and engineering machines that can exhibit human intelligence characteristics. ML, on the other hand, is a prospective AI application based on the premise that machines can learn from enormous amounts of data. Finally, the primary distinction between DL and ML is determined by how each algorithm learns [16]. ML is a subset of AI that focuses on creating models that can learn from previously gathered data and increase their accuracy without being explicitly programmed. Algorithms in ML are fundamentally capable of processing vast amounts of data in order to uncover important patterns for the training phase. When presented with additional input data after an initial training period, an ML model may make correct predictions [16]. In general, any ML workflow can be broken down into five major phases: Problem Definition & Data Collection, Data Preparation, Algorithm Selection, Model Training & Validation, Model Testing.

There are several ways in ML that try to help identify the best solution to the defined problem, these ways are Supervised, Unsupervised, Semi-supervised and

Reinforcement learning, see Fig. 2. [16]. In Machine learning, there are many methods some are traditional methods such as RF, DT, SVM, etc. and DL.

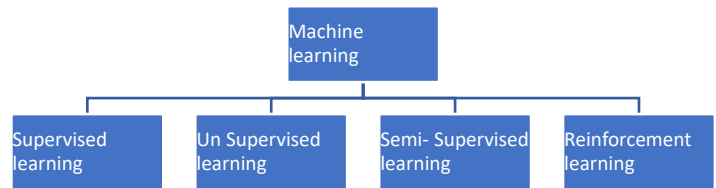


Fig. 2: Machine Learning types [16]

DL is a subcategory of ML that predominantly employs Artificial Neural Networks (ANNs), also known as Feed-Forward Neural Networks (FFNNs), to analyze enormous quantities of data in order to solve complicated tasks that would normally need human intervention. Nowadays, DL algorithms are widely acknowledged as being well-integrated into various goods and services, and they are widely used in a variety of fields ranging from law enforcement to financial services and customer support [16]. Some of DL methods are **Convolution neural network (CNN)**, **Recurrent neural network (RNN)**, **DBN**... etc. See Fig. 3.

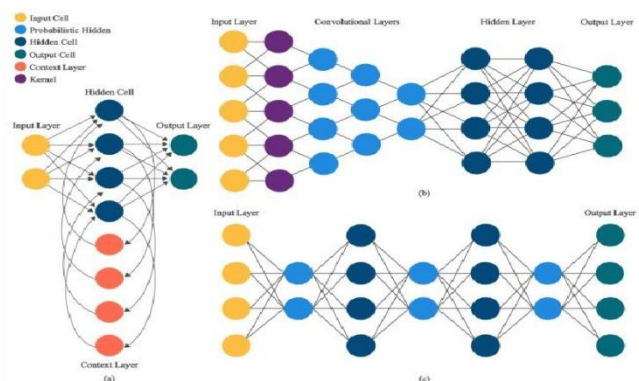


Fig. 3: A Deep learning various Architecture a) RNN b) CNN c) DBN [17]

Network analysis for IDS is divided into three categories: a) signature-based, which is mostly used to detect known attacks by avoiding a large number of false alarm rate (FAR), b) anomaly-based, which is mostly used to identify anomalous network and system behavior, and c) hybrid-based, which is a combination of a) and b). Others have classified the assaults into four primary groups [18]. Denial of service (DOS) is an attack in which a cybercriminal keeps the network system busy or runs out of memory resources, causing the legitimate user's access request to be ignored. A remote to local (R2L) attack occurs when a remote user attempts to get local access through a network by exploiting its weaknesses. A user-to-root (U2R) attack occurs when a legitimate user with limited network access attempts to achieve root privileges. Probing is an assault in which a cyber-criminal scans a computer system

or network in order to exploit weaknesses and vulnerabilities for future exploitation [19]. To combat the rise of new threats, anomaly-based ids utilize machine learning algorithms that first learn from data to monitor network traffic and then determine whether it is benign or malicious traffic [20].

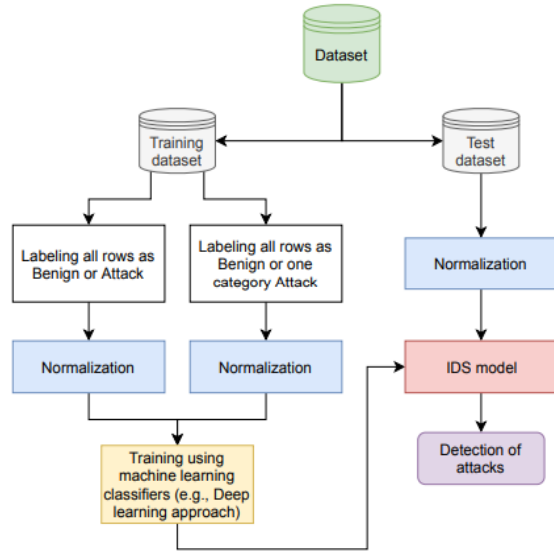


Fig. 4: Intrusion detection system [21]

III. LITERATURE SURVEY

Many studies have been conducted on intrusion detection systems, which are regarded as one of the cybersecurity concerns. This work focuses on the research and implementation of machine learning algorithms on several datasets, including the KDD Cup 99 dataset, the NSL-KDD dataset, the CICIDS-2017 dataset, the CICIDS-2018 dataset, CICAndMal2017 dataset, the ISCX dataset, and the UNSW-NB15 dataset. The primary goal of the data set is to identify attacks. In 2015, Rupam et al. [22] Build a model on the dataset (NSLKDD) using Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO), these algorithms are combined with SVM, and the result is 98.89%. In 2015, Bisyrion et al. [23] Apply the SVM algorithm on the dataset (KDD CUP99), and the outcome was 96.08%. In 2015, Neha et al. [24] performed an experiment using two different classification algorithms on the KDD CUP99 dataset and the experimental results showed that the C4.5 decision tree pruning gives the highest accuracy of 95.09%. In 2015, Zahangir et al. [25] proposed a network-based intrusion detection system based on deep belief, and it was welcomed as an addition in the field of network penetration classification. The DBN-based intrusion detection system had about 97.5% test accuracy after it was applied to the KDD data set using only 40% of the data used for training purposes and fifty iterations. In 2015, Adel et al. [26] Used the ISCX dataset, a DT was utilized, and the outcome was 99 percent. In 2016, Govind et al. [27] Trained the classification-based intrusion detection model and used the NSL-KDD dataset to train the classification-based intrusion detection model, where the accuracy with svm was 37.89 percent, NB was 23.83 percent, RF was 82.35 percent, and the DT was 79.9

percent. They used the KDD dataset only, where the accuracy with svm was 78.84 percent, the NB was 90.68 percent, RF was 88.65% and the DT was 91.13 % respectively. In 2016, Goeschel. [28] carried out an experiment on the KDD CUP99

dataset using several classification methods such as SVM, DT, and NB. The results indicated 99.62 percent accuracy. In

2016, Jasmin et al. [29] Used DT approach on the dataset (KDD), and the result was 86.29%. In 2017, Bhupendra et al. [30] Used DT technique on the dataset (NSL-KDD), and the outcome was 90.3%. In 2017, Donghwoon et al. [31] Utilized DL on the NSL-KDD dataset, with a 90.4% accuracy. In 2017, Prakash et al. [32] made an evaluation of the RF algorithm on NSL KDD and KDD CUP99 data which showed an accuracy of 97.10 percent and 98.10 percent respectively. In 2018, Arif et al. [33] conducted a hybrid technique for a dataset KDD CUP99 in which Particle Swarm Optimization (PSO) was used to clip a node and a truncated DT was utilized to classify network intrusions with 92.87%. In 2018, Preeti et al. [34] employed DT approach to the dataset (KDD), with a result of 98.98%, and the Ensemble of ANN, SVM, and Machine Learning, Automation, Robotics and space (MARS) approach, with a value of 99.85%. In 2019, Ying et al. [35] used Genetic Algorithm (GA) with DBN since it can create optimum people. The DBN can analyze complicated and high-dimensional data efficiently, and the classification results are excellent. The DBN used the acquired network structure as a model for intrusion detection and classification attacks, whereas GA employed the NSL-KDD database to execute numerous iterations to build an ideal network structure. As a result, the problem of how to find an appropriate network design when applying deep learning methods for intrusion detection is solved, and it enhances classification accuracy and model generalization while reducing network complexity, with a classification accuracy of 99.45%. In 2019, ayyaz, et al, [36] the RNN technique was used to train the NSL-KDD data set, and the model accuracy was 94.5 percent.

In 2019, Ying et al. [37] Used RF classification technique on the traffic detection module. To assess detection accuracy, the algorithm was tested on datasets including NSL-KDD. The testing findings demonstrate that the suggested detection algorithm achieved an accuracy rate of 99.95 %percent. In 2019, Yuyang et al. [38] trained a classification-based intrusion detection model using the NSL-KDD dataset, where RF had a 95.1 percent accuracy and the DT had a 93.9 percent accuracy, RF had a 96.3 percent accuracy and the DT had a 95.7 percent accuracy. In 2019, Jonghoon et al. [39] Used SVM on the dataset NSLKDD, and the outcome was 89.7%. In 2019, Vinayakumar et al. [40] trained a classification-based intrusion detection model on the NSL-KDD dataset for classification training-accuracy-dependent intrusion detection model with 75.3 percent RF, 76.3 percent DT, and the KDDCup 99 dataset, with accuracy of 92.5 percent RF, and 92.4 percent DT. In 2019, Ahmed et al. [41] utilized a reduced error pruning (REP) tree on the NSL-KDD data set, and the score was 93.4%. In 2019, Seongrae et al.[42]

examined network data and mentioned that the naked IDS may identify network threats. The goal of this study was to assess the performance of two data mining approaches for identifying network attacks: SVM and FANN. The accuracy of the two methods was computed using the test dataset of KDD to compare their performance. FANN was shown to be more successful than SVM in identifying intrusion as it has achieved 96.7. In 2020, Chaouki et al. [43] used a multi-category data collection for this investigation. The NSL-KDD dataset was evaluated using the SVM method, which had an accuracy of 96.72%, the DT algorithm showed an accuracy of 99.65%, the RF algorithm which had an accuracy of 98.99%, the NSGA2 algorithm with an accuracy of 99.12%. In addition, UNSW-NB15 dataset was evaluated on NB algorithm with accuracy of 82.07%, ANN algorithm with precision of 81.34%, c4.5 algorithms with precision of 81.42%, DT algorithm with accuracy of 94.90, NB Tree algorithm with precision 66.00.c. Moreover, CICIDS2017 dataset was evaluated using RF algorithm with precision of 97.00%, decision tree algorithm with precision of 87.95, NB algorithm with precision of 4.00, Multi-layer perceptron (MLP) algorithm with precision of 83.00%, c4.5 algorithms with precision of 85.32%, and ID3 algorithm with precision of 98.00. In 2020, Mohamed et al. [44] The CIC-IDS2018 dataset was utilized and tested using DNN, and the accuracy was 97.281 percent, 97.310 percent with RNN, and 97.376 percent with CNN. In 2020, Zhu et al. [45] used The Android Malware Dataset and applied it to the Long Short Term Memory (LSTM) algorithm, with a 97.22 percent accuracy. In 2021, Abdul Salam et al. [46] trained and tested on NSL-KDD standard data. They examined eight learning algorithms using XGBoost in this post. They evaluated them using a number of grading scales to determine the advantages and disadvantages of using one or more. The proposed method focused on detecting anomalies and protecting the Specially Designated Nationals (SDN) platform from attacks in real-time scenarios, where the accuracy of using j48 was 81.05%, NB with 76.56%, NB Tree with 82.02%, RF with 80.67%, Random Tree(RT) with an a81.59%, Multi-layer perceptron(MLP) with 77.41%, SVM with 69.52, DNN is 75.75% and proposed method(XGboost) with an accuracy of 95.55%. In 2021, Omar et al. [47] utilized The data set CICAndMal2017, where the accuracy of the DT method was 99.3, while the accuracy of the RF approach was 97.02. The accuracy of the k-Nearest Neighbor (k-NN) approach was 98.16, the accuracy of the XGBoost technique was 99.20, and the accuracy of the MLP algorithm was 91.64. In 2022 Mohammed et al.[48] used Partial Decision Tree Technology (PART)- a lightweight classifier that distinguishes DDoS network patterns from normal traffic with a detection accuracy of over 99.77%. The classifier proposed in CICIDS2017 was trained using a number of properties and verified using the CICDDoS2019 dataset.

Table I shows a comparison between the algorithms used with the data set and the accuracy of each algorithm on the data set.

Table I: The accuracy of classification method

Dataset	Reference	Year	Classification Methods	Accuracy
NSL-KDD	[22]	2015	Aco +Pso+ SVM	98.89
KDD-CUP99	[23]	2015	SVM	96.08
KDD-CUP99	[24]	2015	C4.5 dt with pruning	95.09
KDD	[25]	2015	DBN	97.5
ISCX	[26]	2015	DT	99
NSL-KDD	[27]	2016	SVM	37.89
NSL-KDD	[27]	2016	NB	23.83
NSL-KDD	[27]	2016	RF	82.35
NSL-KDD	[27]	2016	DT	79.91
KDD	[27]	2016	SVM	78.84
KDD	[27]	2016	NB	90.68
KDD	[27]	2016	RF	88.65
KDD	[27]	2016	DT	91.13
KDD-CUP99	[28]	2016	SVM and DT and NB	99.62
KDD	[29]	2016	DT	86.29
NSL-KDD	[30]	2017	DT	90.3
NSL-KDD	[31]	2017	DBN	90.4
NSL-KDD	[32]	2017	RF	97.1
KDD-CUP99	[32]	2017	RF	98.1
KDD-CUP99	[33]	2018	PSO and DT	92.87
KDD	[34]	2018	DT	98.98
KDD	[34]	2018	ANN, SVM and MARS	99.85
NSL-KDD	[35]	2019	DBN	99.45
NSL-KDD	[36]	2019	RNN	94.5
NSL-KDD	[37]	2019	RF	99.95
NSL-KDD	[38]	2019	DT	93.9
NSL-KDD	[38]	2019	RF	95.1
KDD-CUP99	[38]	2019	RF	96.3
KDD-CUP99	[38]	2019	DT	95.7
NSL-KDD	[39]	2019	SVM	89.7
NSL-KDD	[40]	2019	RF	75.3
NSL-KDD	[40]	2019	DT	76.3
KDD-CUP99	[40]	2019	RF	92.5
KDD-CUP99	[40]	2019	DT	92.4
NSL-KDD	[41]	2019	REP Tree	93.4
KDD	[42]	2019	FANN and SVM	96.7
NSL-KDD	[43]	2020	NSGA2	99.12
NSL-KDD	[43]	2020	SVM	96.72
NSL-KDD	[43]	2020	DT	99.65
NSL-KDD	[43]	2020	RF	98.99
UNSW-NB15	[43]	2020	DT	94.9
UNSW-NB15	[43]	2020	NB	82.07
UNSW-NB15	[43]	2020	ANN	81.34
UNSW-NB15	[43]	2020	C4.5	81.42
UNSW-NB15	[43]	2020	NB Tree	66
CICIDS2017	[43]	2020	RF	97
CICIDS2017	[43]	2020	DT	87.95
CICIDS2017	[43]	2020	NB	4
CICIDS2017	[43]	2020	MLP	83
CICIDS2017	[43]	2020	C4.5	85.32
CICIDS2017	[43]	2020	ID3	98
CICIDS2018	[44]	2020	DNN	97.281
CICIDS2018	[44]	2020	RNN	97.31
CICIDS2018	[44]	2020	CNN	97.376
Android Malware	[45]	2020	LSTM	97.22
NSL-KDD	[46]	2020	J48	81.05
NSL-KDD	[46]	2020	NB	76.56
NSL-KDD	[46]	2020	NB Tree	82.02
NSL-KDD	[46]	2020	RF	80.67
NSL-KDD	[46]	2020	RT	81.59
NSL-KDD	[46]	2020	MLP	77.41

NSL-KDD	[46]	2020	SVM	69.52
NSL-KDD	[46]	2020	DNN	75.75
NSL-KDD	[46]	2020	XGBost	95.55
CIC AND MAL2017	[47]	2021	DT	99.3
CIC AND MAL2017	[47]	2021	RF	97.02
CIC AND MAL2017	[47]	2021	K-NN	98.16
CIC AND MAL2017	[47]	2021	DT	98.98
CIC AND MAL2017	[47]	2021	XGB	99.2
CIC AND MAL2017	[47]	2021	MLP	91.64
CIC AND MAL2017 and CICDDoS 2019	[48]	2022	PART	99.77

IV. Conclusion

Cybersecurity has become a global concern with improvements in security measures to detect and respond to cyber-attacks. The previously used traditional security systems are no longer sufficient because these systems are incompetent in detecting previously invisible and polymorphic attacks.

Machine learning techniques play a vital role in many applications of the cybersecurity paradigm. The paper revealed a rapidly growing interest in machine learning and cybersecurity in our world. In this paper, machine learning techniques and threats to computer networks have been examined by presenting a comprehensive survey that presents a literature review on machine learning techniques for intrusion detection.

The applications of machine learning models in the field of cybersecurity which are attacks based on various criteria such as detection rate, time complexity, and classification time have been briefly presented. This paper detects new attacks, zero attacks, and accuracy of the ML model. A data set is absolutely essential for training and testing ML models. Concerning the previous studies mentioned above, it has been concluded that intrusion can be detected. For example, in the case of using NSL-KDD data sets, the RF classifier achieved the best accuracy of 99.95%. In KDD Cup 99, the best classifier 99.62% was a combination of SVM, DT and NB, for the CICIDS-2017 data set, the best classifier was ID3, thus achieving an accuracy of 98%. As for the CICIDS-2018 data set, the best classifier was CNN, and therefore achieved 97.376%. According to the ISCX data set, the best classifier was DT, getting 99%, but the CICAndMal2017 data set was better. The accuracy obtained from the DT classifier is 99.3. With regard to the UNSW-NB15 dataset, the best accuracy resulting from the DT algorithm was 94.9, all of which used security datasets.

Finally, it has been found that the safe use of machine learning techniques in cyberspace offers some guarantees of high-level validity, speed and accuracy for the model. Some of the important machine learning techniques, data sets, and accuracy in cybersecurity as well as an extensive bibliography have been provided briefly in the field.

References

1. L. Xu, W. He, and S. Li, "Internet of Things in industries: a survey," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2233-2243, 2014.
2. D. K. Bhattacharyya and J. K. Kalita, *Network anomaly detection: A machine learning perspective*. Chapman and Hall/CRC, 2013.
3. V. Ambalavanan, "Cyber Threats Detection and Mitigation Using Machine Learning," in *Handbook of Research on Machine and Deep Learning Applications for Cyber Security*: IGI Global, pp. 132- 149, 2020.
4. T. Thomas, A. P. Vijayaraghavan, and S. Emmanuel, "Machine Learning and Cybersecurity," in *Machine Learning Approaches in Cyber Security Analytics*: Springer, pp. 37-47, 2020.
5. S. Dey, Q. Ye, S. Sampalli, A machine learning based intrusion detection scheme for data fusion in mobile clouds involving heterogeneous client networks, *Information Fusion*, Volume 49, pp 205-215, 2019.
6. B. Geluvaraj, P. M. Satwik and T. A. Ashok Kumar. The future of cybersecurity: Major role of artificial intelligence, machine learning, and deep learning in cyberspace. In *International Conference on Computer Networks and Communication Technologies*; Springer, pp. 739-747, 2018.
7. P. Jain, *Machine Learning Versus Deep Learning for Malware Detection*. Master 's Thesis, San Jose State University, USA, 2019.
8. Rao, R.S.; Pais, A.R. Detection of phishing websites using an efficient feature-based machine learning framework. *Neural Comput and Applic*, Volume 31, pp 3851-3873, 2019.
9. M. Alauthman, A. Almomani, M. Alweshah, W. Omoushd, K. Alieyane. Machine Learning for phishing Detection and Mitigation. *Mach. Learn. Comput. Cyber Secur. Princ. Algorithms and Pract*, Volume 26, pp 48-74, 2019.
10. A. A. Alurkar, S.B.Ranade, S.V. Joshi, S.S. Ranade, G.R.Shinde, P.A Sonewar, P.N. Mahalle, A Comparative Analysis and Discussion of Email Spam Classification Methods Using Machine Learning Techniques. In *Applied Machine Learning for Smart Data Analysis*; CRC Press: Boca Raton, FL, USA, pp. 185, 2019.
11. E.G.Dada, J.S.Bassi, H.Chiroma, A.O. Adetunmbi, O.E. Ajibuwa, Machine learning for email spam filtering: Review, approaches and open research problems. *Heliyon*, Volume 5, 2019.
12. H.A.Shukur, S. Kurnaz, Credit Card Fraud Detection Using Machine Learning Methodology. *Int. J. Comput. Sci. Mob. Comput.* Volume 8, pp 257-260, 2019.
13. V.Jusas, S.Japertas, T. Baksys, S. Bhandari, Logical filter approach for early stage cyber-attack detection. *Comput. Sci. Inf. Syst.*, Volume 16, pp 491-514, 2019.
14. J. Zhao, Q. Yan, J. Li, M. Shao, Z. He, B. Li, TIMiner: Automatically extracting and analyzing categorized cyber threat intelligence from social data, *Computers & Security*, Volume 95, pp 101867, 2020.
15. J.Zhang, Distributed network security framework of energy internet based on internet of things, *Sustainable Energy Technologies and Assessments*, Volume 44, pp 101051, 2021.
16. E. Sula, SecRiskAI: A Machine Learning-based Tool for Cybersecurity Risk Assessment, Department of Informatics (IFI), 2021.
17. K. Shaukat, S. Luo, V. Varadharajan, I. A. Hameed and M. Xu, "A Survey on Machine Learning Techniques for Cyber Security in the Last Decade," in *IEEE Access*, vol. 8, pp. 222310-222354, 2020, doi: 10.1109/ACCESS.2020.3041951.
18. J. Raiyn, "A survey of cyber-attack detection strategies," *International Journal of Security and Its Applications*, vol. 8, no. 1, pp. 247-256, 2014.
19. P. Ganapathi, "A Review of Machine Learning Methods Applied for Handling Zero-Day Attacks in the Cloud Environment," in *Handbook of Research on Machine and Deep Learning Applications for Cyber Security*: IGI Global, pp. 364-387, 2020.
20. L. A. H. Ahmed and Y. A. M. Hamad, "Machine Learning Techniques for Network-based Intrusion Detection System: A Survey Paper," 2021 National Computing Colleges Conference (NCCC), pp. 1-7, 2021, doi: 10.1109/NCCC49330.2021.9428827.
21. M. Ferrag and L. Maglaras, "Delivery Coin: An IDS and Blockchain-Based Delivery Framework for Drone-Delivered Services", *Computers*, vol. 8, no. 3, p. 58, 2019.

22. R. K. Sharma, H. K. Kalita, P. Borah, Analysis of machine learning techniques based intrusion detection systems. In Proceedings of the 3rd International Conference on Advanced Computing, Networking and Informatics; Springer, pp. 485–493, 2015.
23. B. W. Masduki, K. Ramli, F. A. Saputra and D. Sugiarto, "Study on implementation of machine learning methods combination for improving attacks detection accuracy on Intrusion Detection System (IDS)," 2015 International Conference on Quality in Research (QiR), pp. 56–64, 2015, doi: 10.1109/QiR.2015.7374895.
24. N. G. Relan and D. R. Patil, "Implementation of network intrusion detection system using variant of decision tree algorithm," 2015 International Conference on Nascent Technologies in the Engineering Field (ICNTE), pp. 1–5, 2015, doi:10.1109/ICNTE.2015.7029925.
25. Z. Alom, V.R. Bontupalli, T. M. Taha. "Intrusion Detection Using Deep Belief Network and Extreme Learning Machine in Proceedings of the 2015 National Aerospace and Electronics Conference (NAECON), Volume 3, pp 35–56,2015.
26. A. Ammar, A decision tree classifier for intrusion detection priority tagging. J.Comput. Commun. Volume 3, pp 52,2015.
27. G.P. Gupta, M. Kulariya, A Framework for Fast and Efficient Cyber Security Network Intrusion Detection Using Apache Spark, Procedia Computer Science, Volume 93, pp 824–831,2016.
28. K.Goeschel, Reducing false positives in intrusion detection systems using data-mining techniques utilizing support vector machines, decision trees, and naive Bayes for off-line analysis. In Proceedings of the SoutheastCon 2016, Norfolk, VA, USA, pp. 1–6, 2016.
29. J.Kevric, S.Jukic, A. Subasi, "An effective combining classifier approach using tree algorithms for network intrusion detection." Neural Computing and Applications, Volume 28, pp 1051–1058,2016.
30. B.Ingre, A.Yadav, A.K. Soni, Decision tree based intrusion detection system for NSL-KDD dataset. In Proceedings of the International Conference on Information and Communication Technology for Intelligent Systems, Ahmedabad, India, pp. 207–218,2017.
31. D. Kwon, H. Kim, J. Kim, S.C. Suh, I. Kim, K.J. Kim, A survey of deep learning-based network anomaly detection. Clust. Comput, Volume 22, pp 949–961,2017.
32. P.Chandra, U.K. Lilhore, N.Agrawal, Network intrusion detection system based on modified Random forest classifiers for kdd cup-99 and nsl-kdd Dataset. Int. Res. J. Eng. Technol, Volume 4, pp 786–791, 2017.
33. A.J.Malik, F.A.Khan, A hybrid technique using binary particle swarm optimization and decision tree pruning for network intrusion detection. Clust. Comput, Volume 21, pp 667–680, 2018.
34. P. Mishra, V.Varadharajan, U.Tupakula, E.S.Pilli, Tutorials. A detailed investigation and analysis of using machine learning techniques for intrusion detection. IEEE Commun. Surv. Tutor, Volume 21, pp 686–728 ,2018.
35. Y. Zhang, P. Li, X.J.I.A.Wang., Intrusion detection for IoT based on improved genetic algorithm and deep belief network. IEEE Access, Volume 7, pp 31711–31722, 2019.
36. A.U.H.Qureshi, H. Larijani, N. Mtetwa, A. Javed, J.J.C. Ahmad, RNN-ABC: A New Swarm Optimization Based Technique for Anomaly Detection. Computers, Volume 8, pp 59, 2019.
37. Y.Gao, H.Wu, B.Song, Y.Jin, X. Luo, X.J.I.A. Zeng, A Distributed Network Intrusion Detection System for Distributed Denial of Service Attacks in Vehicular Ad Hoc Network. IEEE Access, Volume 7, pp 154560–154571. 2019.
38. Y.Zhou, G. Cheng, S. Jiang, M. Dai, Building An Efficient Network Intrusion Detection System Based on Feature Selection and Ensemble Classifier. Volume arXiv , pp 1904.01352, 2019.
39. J. Lee, J. Kim, I.Kim, K.Han, Cyber Threat Detection Based on Artificial Neural Networks Using Event Profiles. IEEE Access, Volume 7, pp 165607–165626, 2019.
40. R.Vinayakumar, M. Alazab, K. Soman, P. Poornachandran, A. Al-Nemrat, S.J.I.A. Venkatraman, Deep Learning Approach for Intelligent Intrusion Detection System. IEEE Access, Volume 7, pp 41525–41550, 2019.
41. A.Ahmim, L. Maglaras, M.A Ferrag, M. Derdour, H. Janicke, A novel hierarchical intrusion detection system based on decision tree and rules-based models. In Proceedings of the 2019 15th International Conference on Distributed Computing in Sensor Systems (DCOSS), Santorini Island, pp. 228–233, 2019.
42. S.Jo , H.Sung, B. Ahn, A comparative study on the performance of intrusion detection using decision tree and artificial neural network models. J. Korea Soc. Digit. Ind. Inf. Manag, Volume 11, pp 33–45, 2019.
43. C. Khammassi, S. Krichen, A NSGA2-LR wrapper approach for feature selection in network intrusion detection, Comput. Netw. Volume 172, 107183, 2020.
44. M.A. Ferrag, L. Maglaras, S. Moschouiannis, H. Janicke, Deep learning for cyber security intrusion detection: approaches, datasets, and comparative study, J. Inf. Secur. Appl, Volume 50, 2020
45. Z. Ma, H. Ge, Z. Wang, Y. Liu, and X. Liu, "Droidetec: Android malware detection and malicious code localization through deep learning," arXiv preprint arXiv:2002.03594, 2020.
46. A.O.Alzahrani, M.J.F. Alenazi, "Designing a Network Intrusion Detection System Based on Machine Learning for Software Defined Networks" Future Internet Volume 13, pp 111, 2021. <https://doi.org/10.3390/fi13050111>.
47. Omar Shamil Ahmed, Omar Abdulmunem Ibrahim Al-Dabbagh, "Ransomware Detection System Based on Machine Learning," Journal of Education and Science, vol. 30, pp. 86–102, 2021.
48. M. I. Kareem and M. N. Jasim, "DDoS Attack Detection Using Lightweight Partial Decision Tree algorithm," 2022 International Conference on Computer Science and Software Engineering (CSASE), pp. 362–367, 2022. doi: 10.1109/CSASE51777.2022.9759824.